METRIC

MIL-DTL-32656

2 June 2020

DETAIL SPECIFICATION

ARC FAULT PROTECTION SYSTEM, GENERAL SPECIFICATION FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the Arc Fault Protection System (AFPS) that detects an arc fault in electrical equipment such as switchboards, load centers, motor drives, and produces a signal to automatically trip upstream circuit breakers to isolate the affected section of the enclosure. High resistance arcing faults can cause damage to power distribution systems since the inherent operational characteristics of Navy circuit breakers may not isolate the affected enclosures for several seconds due to relatively low current levels associated with this type of arcing fault. The AFPS detects high impedance arcs in series with system loads and phase-to-phase arcs and has the capability to command a breaker trip signal within the required time period.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

COMMERCIAL ITEM DESCRIPTIONS

A-A-59486	-	Padlock Set (Individually Keyed or Keyed Alike)
A-A-59487	-	Padlock (Key Operated)

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-901	-	Shock Tests, H.I. (High-Impact) Shipboard Machinery, Equipment, and Systems, Requirements for
MIL-DTL-917	-	Electric Power Equipment, Basic Requirements for
MIL-DTL-2036	-	Enclosures for Electric and Electronic Equipment, Naval Shipboard
MIL-DTL-15024	-	Plates and Tags for Identification of Equipment, General Specification for
MIL-P-15024/5	-	Plates, Identification

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 5975

DISTRIBUTION STATEMENT A. Approved for public release. Distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:29Z
Check the source to verify that this is the current version before use.

MIL-DTL-32656/1	-	Arc Fault Protection System, Photosensor
MIL-DTL-32656/2	-	Arc Fault Protection System, Continuous Thermal Monitoring Device, Thermal Ionization Type
MIL-PRF-39007/10	-	Resistor, Fixed, Wirewound (Power Type), Nonestablished Reliability, Established Reliability, and Space Level Style RWR84
MIL-PRF-39007/11	-	Resistor, Fixed, Wirewound (Power Type), Nonestablished Reliability, Established Reliability, and Space Level Style RWR89
MIL-PRF-85045	-	Cables, Fiber Optic, General Specification for

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-108	-	Definitions of and Basic Requirements for Enclosures for Electric and Electronic Equipment
MIL-STD-167-1	-	Mechanical Vibrations of Shipboard Equipment (Type I – Environmental and Type II – Internally Excited)
MIL-STD-202-103	-	Method 103, Humidity (Steady State)
MIL-STD-202-301	-	Method 301, Dielectric Withstand Voltage
MIL-STD-202-302	-	Method 302, Insulation Resistance
MIL-STD-461	-	Requirements for the Control of Electromagnetic Interference Characteristics of Subsystems and Equipment
MIL-STD-810	-	Environmental Engineering Considerations and Laboratory Tests
MIL-STD-1310	-	Shipboard Bonding, Grounding, and Other Techniques for Electromagnetic Compatibility, Electromagnetic Pulse (EMP) Mitigation, and Safety
DOD-STD-1399-70-1	-	Interface Standard for Shipboard Systems, Section 070 – Part 1, D.C. Magnetic Field Environment (Metric)
MIL-STD-1399-300-1	-	Low Voltage Electric Power, Alternating Current
DOD-STD-1399-301	-	Interface Standard for Shipboard Systems, Section 301A, Ship Motion and Attitude (Metric)
MIL-STD-1686	-	Electrostatic Discharge Control Program for Protection of Electrical and Electronic Parts, Assemblies and Equipment (Excluding Electrically Initiated Explosive Devices)

(Copies of these documents are available online at https://quicksearch.dla.mil/.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

IPC

IPC-CC-830	-	Qualification and Performance of Electrical Insulating Compound for Printed Wiring Assemblies

(Copies of this document are available online at www.ipc.org.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:29Z
Check the source to verify that this is the current version before use.

TELECOMMUNICATIONS INDUSTRY ASSOCIATION (TIA)

TIA-485 - Electrical Characteristics of Generators and Receivers for Use in Balanced Digital Multipoint Systems

(Copies of this document are available online at www.tiaonline.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein (except for related specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheet. In the event of any conflict between the requirements of this specification and the specification sheet, the latter shall govern.

3.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.3 Materials of construction. Materials shall be in accordance with MIL-DTL-917 unless otherwise approved by the Technical Authority (see 6.2 and 6.3.2).

3.3.1 Conformal coat. All circuit boards shall be conformal coated in accordance with IPC-CC-830.

3.3.2 Creepage and clearance distances. Creepage and clearance distances shall be in accordance with MIL-DTL-917 unless otherwise approved by the Technical Authority (see 6.2).

3.3.3 Fiber optic cable. Fiber optic cable shall be in accordance with MIL-PRF-85045. Fiber optic sensors shall use optical grade fiber optic cord when required by the design.

3.3.4 Electrical connectors. Electrical connectors shall be in accordance with MIL-DTL-917 unless otherwise approved by the Technical Authority (see 6.2).

3.3.5 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.4 Arc fault protection system (AFPS) components. Unless otherwise specified (see 6.2), the AFPS shall consist of a control unit, photosensors, continuous thermal monitoring device(s) (CTMD), and interface components when necessary for system operation.

a. Control unit. The control unit shall reside in an electrical enclosure that contains control components required to receive signals from photosensors and trips circuit breakers via external relays and receive signals from CTMDs and actuate alarms, log events, and provide a user interface.

b. Photosensors. Photosensors shall be point sensors or loop type sensors.

c. CTMDs. Unless otherwise specified (see 6.2), CTMDs shall detect off-gassing of insulating materials to indicate the temperature of components.

d. Interface components. Interface components shall include cables, terminal boards, junction boxes, sensor interrogation devices (if used), and communication devices.

3.4.1 Control unit.

3.4.1.1 Control unit enclosure. The control unit enclosure shall meet the construction requirements of MIL-DTL-2036. The degree of enclosure shall be spraytight as defined by MIL-STD-108.

3.4.1.2 Physical. The control unit shall be designed for and include provisions for mounting to a vertical flat surface.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:29Z
Check the source to verify that this is the current version before use.

3.4.1.2.1 Grounding. All external parts capable of electrical conduction shall be bonded and at the same electrical potential at all times. Provisions shall be made to allow shipboard bonding to ground potential in accordance with MIL-STD-1310.

3.4.1.2.2 Data ports. The control unit shall have no externally accessible data ports or wireless connections. All unused or unnecessary ports inside enclosures shall be capable of being disabled physically or logically. Physical access to the ports shall require opening enclosure access panels and covers. Any data ports that will allow for the modification of existing software or installation of new software on the device shall be located within the secured enclosure.

3.4.1.2.3 Enclosure access panels or covers. When provided, all enclosure access panels or covers shall utilize lead-wire type or other tamper-evident seals and padlocks in accordance with A-A-59486 or A-A-59487

3.4.1.2.4 Connector caps. Connector caps shall be provided for unused external connectors. Connector caps shall be electrically conductive.

3.4.1.3 Temperature ranges. The temperature ranges for the control unit shall be as follows:

3.4.1.3.1 Operating temperature range. The control unit shall operate in a temperature range of 0 to +50 °C.

3.4.1.3.2 Non-operating temperature range. The control unit shall operate after storage in a temperature range of -40 to +75 °C.

3.4.1.4 Nameplates. Manufacturer identification, other than that allowed for the nameplates, shall not appear on the equipment. Markings shall be in accordance with MIL-DTL-15024 and MIL-P-15024/5.

3.4.1.4.1 Electrostatic discharge (ESD) protection requirements. When specific parts, modules, connectors/receptacles, or subassemblies sensitive to damage by ESD are used, the devices shall be clearly marked with ESD labeling in accordance with MIL-STD-1686.

3.4.1.5 Arc fault response. The control unit shall detect arcing within electrical enclosures and furnish both local (control unit) and remote (external) alarms. Upon detection of an arc, the AFPS shall drive the input of an existing auxiliary relay to open the appropriate circuit breakers to isolate the affected section of the enclosure and curtail the arc in time to minimize damage. The AFPS shall continuously monitor its photosensors for any activity unique to an arc fault event. The time required to validate an arc event shall begin as soon as any one sensor output crosses the threshold voltage and shall not reset until all sensors lose the light signal. Unless otherwise specified (see 6.2), should an arc fault event signal greater than 45 milliseconds in duration be detected, the control unit, within 105 milliseconds, shall activate the associated trip circuit for approximately 2 seconds and issue an audible alarm. The trip signal shall be released after approximately 2 seconds. The display shall indicate the sensor location until the alarm is cleared. The alarm shall be capable of being both acknowledged and cleared via the control unit human machine interface (HMI). The photosensor sampling period shall be fast enough to support detection of events less than 2 milliseconds in duration.

3.4.1.6 Thermal response. The control unit shall receive inputs for over-temperature events and shall generate an audible local alarm. The over-temperature alarm limit and accuracy shall be in accordance with the applicable specification sheet. No breakers shall be tripped as a result of an alarm. The display shall indicate the sensor location until the alarm is cleared. The alarm shall be capable of being both acknowledged and cleared via the control unit.

3.4.1.7 Power. The control unit shall operate in accordance with MIL-STD-1399-300-1, type I power at 115 volts alternating current (VAC). The control unit shall supply power to all connected sensors and trip relay coils.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:29Z
Check the source to verify that this is the current version before use.

3.4.1.8 Control unit features.

3.4.1.8.1 Built-in test (BIT). The control unit shall be capable of automatically executing the BIT without operator action on a frequency of at least once every 24 hours, as well as be capable of execution upon initiation by the operator from the HMI. The BIT shall include end-to-end testing of the control unit, the sensors, and the coil of the remote relay used to trip the circuit breaker. The BIT shall indicate a failure if the remote trip interface relay circuit is shorted or open. The BIT shall be verified to indicate a failure if any sensor circuit is open, shorted, or the sensor is otherwise nonfunctional. For photo-sensing mechanisms, the system shall contain a permanent light source that is activated by the BIT. BIT activation shall cause activation of a light source sufficient to ensure the detection of the photo-sensing mechanism with minimal degradation being exercised. In no case shall activation of the BIT cause activation of remote trip interface relays or alarm outputs. A BIT shall be required for CTMDs that detect off-gassing of insulating materials to ensure the detection mechanism for the over-temperature event is exercised. BIT testing shall manipulate the CTMDs such that the alarm and full clean air operation ranges are verified. If any component fails the BIT during control unit initialization (boot-up) or any other BIT test, that component shall be identified and not excluded from any future BIT tests.

3.4.1.8.2 HMI. The control unit shall have an operator interface to allow for configuration, operation, and diagnostics of the system. The control unit shall be fully commandable from the interface to display system status, silence and clear alarms, perform the BIT (if required), and to perform maintenance and system configuration functions.

3.4.1.8.3 Alarms.

3.4.1.8.3.1 Audible arc response alarm. The control unit shall activate an audible local alarm that sounds when a validated arcing fault is detected by the system. The alarm shall be capable of being reset from the control unit HMI. The alarm shall be capable of being silenced without reset from the control unit HMI. The alarm shall not be capable of being reset from an external device or over the supervisory network.

3.4.1.8.3.2 Audible thermal response alarm. Unless otherwise specified (see 6.2), the control unit shall activate an audible local alarm that sounds when an over-temperature event is detected by the CTMD sensors. The alarm shall be capable of being reset from the control unit HMI. The alarm shall be capable of being silenced without reset from the control unit HMI. The alarm shall not be capable of being reset from an external device or over the supervisory network.

3.4.1.8.3.3 Remote arc response alarm. The control unit shall activate a remote alarm circuit when a validated arcing fault is detected by the system. Each alarm interface shall provide for the connection of a 20-24 American Wire Gauge (AWG) pair. In the alarm condition, the control unit shall provide for shorting of the alarm pair through contact closure. A 7,000 ohm, ±5 percent, ½- to 5-watt current limiting resistor, or a type RWR89 (in accordance with MIL-PRF-39007/11) or RWR84 (in accordance with MIL-PRF-39007/10) 6,180-ohm resistor shall be installed across the alarm lines such that with alarm contacts open (in the non-alarm condition), the alarm circuit is completed through the resistor to allow for remote alarm supervisory function. Contacts shall be rated not less than 10 volts direct current (VDC).

3.4.1.8.3.4 Remote thermal response alarm. The control unit shall activate a remote alarm circuit when an over-temperature event is detected by the CTMD. Each alarm interface shall provide for the connection of a 20-24 AWG pair. In the alarm condition, the control unit shall provide for shorting of the alarm pair through contact closure. A 7,000 ohm, ±5 percent, ½- to 5-watt current limiting resistor, or a type RWR89 (in accordance with MIL-PRF-39007/11) or RWR84 (in accordance with MIL-R-39007/10) 6,180-ohm resistor shall be installed across the alarm lines such that with alarm contacts open (in the non-alarm condition), the alarm circuit is completed through the resistor to allow for remote alarm supervisory function. Contacts shall be rated not less than 10 VDC.

3.4.1.8.4 Data log/archive file. The control unit shall archive/log data. Data shall include system configuration data, BIT operation, BIT failure data, sensor failure data, arc fault and CTMD (when specified [see 6.2]) event data, and sensor(s) reporting data and location. All data shall be date- and time-stamped. Unless otherwise specified (see 6.2), the data log shall be capable of storing at minimum the most recent 2,000 operating hours). The data log file shall be retained upon a loss of power.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:29Z
Check the source to verify that this is the current version before use.

3.4.1.8.5 Trip circuit. Each trip circuit shall be capable of providing a minimum of 113 milliamps at 115 VAC at a frequency of 60 hertz.

3.4.1.8.6 Embedded software. Equipment furnished with software that is used in the AFPS shall be identified by a nameplate with the name or part number or both, and the version number or designator.

3.4.1.8.7 Modes of operation. The normal mode for operation shall be MONITOR. A mode shall be provided for setup and check called MAINTENANCE. When not in MONITOR mode, the control unit shall be in MAINTENANCE mode, during which time no alarms are reported and no trip relays can be activated. MAINTENANCE mode shall allow for the setup, configuration, replacement, and adjustment of components and component parameters. MAINTENANCE mode shall not allow modifications to the software or operating system, and shall not include an ability to cause power to trip. The selection of MAINTENANCE mode from MONITOR mode shall only be achievable via the local control unit HMI.

a. A remote alarm connection equivalent to the alarm requirements of 3.4.1.8.3.3 shall be provided and shall be activated upon user entry to MAINTENANCE mode. The alarm shall activate upon the first failed attempt at password entry and shall remain active until deactivated from within MAINTENANCE mode (after proper passcode entry). The HMI shall not display any indication of alarm, nor shall the audible alarm sound.

b. The passcode to enter MAINTENANCE mode shall consist of five digits and shall be capable of being changed from within MAINTENANCE mode. Password recovery shall be possible through direct connection of a laptop to the internal data port.

3.4.1.8.7.1 Sensor shutdown. Provisions shall be made to deactivate and to reactivate individual sensors or a group of sensors to isolate a specific enclosure when a control unit has the ability to monitor multiple compartments.

3.4.1.8.8 Cybersecurity (see 6.3.1). When specified (see 6.2), the control unit shall be able to remotely communicate with an external device. The communication protocol and functions shall be as specified (see 6.2).

a. Communication shall be through a physical connection. Wireless communication shall not be permitted. Any components with wireless capability shall be disabled through physical or logical means.

b. Controls shall not be reprogrammable over the connected supervisory network. Controls shall be reprogrammable only via a direct local physical connection inside the control enclosure or HMI access.

c. Communication from control unit to External Control System (ECS) shall include a source identifier.

d. Control unit to ECS communication shall include a method for ECS to confirm the integrity of messages.

e. The control unit shall report the results of integrity checks to the HMI and the log file.

f. The control unit shall continuously maintain state information (such as; previous “n” processes, initial failure exception, stack) necessary to determine the cause of failure.

g. The control unit shall provide for resource exhaustion and have protection against it.

h. The control unit shall provide system audits for indications of software changes from the baseline and attempts at unauthorized access over the supervisory network. Audit completion shall flow to the log file. In the event of audit failure, failure information shall flow to the control unit HMI.

i. The control unit shall be designed with the capability to export audit data for use by separate forensic tools to support investigations in the event of a security incident. Audit data shall consist at minimum of a date/time stamp, the type of activity detected, and the port on which it was detected.

j. The control unit shall protect audit data information and audit tools from unauthorized access, modification, and deletion.

k. Human user authentication shall be performed via user name and password combination.

l. A five-digit numeric passcode shall be used to access MAINTENANCE mode. Passwords for access to software and the operating system through an internal data port shall:

(1) Contain a minimum of 14 characters.

(2) Contain at least one lower-case character.

(3) Contain at least one upper-case character.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:29Z
Check the source to verify that this is the current version before use.

(4) Contain at least one numeric character.

(5) Contain at least one special character.

m. Software and firmware applications shall enforce the limit of five consecutive invalid log-in attempts by a user during a 15-minute period.

n. Authentication parameters shall be configurable to allow for changes.

o. Session authentication (direct or networked) shall utilize a secure communication method that prevents the disclosure of the authentication method.

p. Communication sessions shall expire after a 15-minute period of inactivity.

q. Configuration change privileges for a given level of access to the configuration interface shall be limited to the minimum necessary for the operator of the equipment.

r. The control unit shall fail to a known secure state.

s. The control unit shall recover securely from failures.

t. Alarm outputs shall be one-way from the control unit to an external alarm system. No incoming communication over the alarm ports shall be accepted by the control unit.

u. The control unit control functionality shall be prohibited from utilizing a real-time clock. A real-time clock may be used to support data acquisition and log files as otherwise specified herein, except that the real-time clock shall be logically isolated from control functionality via one-way communications from control functions to data acquisition.

v. The communication between the control unit and any external interface (other than the sensor networks) shall be one-way, output only from the control unit while in MONITOR mode.

w. The control unit shall perform a system reboot while shifting from MAINTENANCE to MONITOR mode.

3.4.1.8.9 Input power switch. A switch mounted on the outside of the control unit shall disconnect input control power. The switch shall have a method of protection from inadvertent contact.

3.4.1.8.10 Input power fuses. Fuses mounted on the outside of the control unit shall provide protection for the input control power lines.

3.4.1.8.11 Restoring the control unit and sensors. Procedures to restore the control unit and its associated sensors to software baseline shall be provided as specified (see 6.2).

3.4.1.8.12 Baseline configurations. Baseline configurations shall be documented. Baseline configurations include but are not limited to network topology, the logical placement of the components, and information about components such as software packages, version numbers and patch information on operating systems and application, and configuration setting and parameters.

3.4.1.8.13 Minimum number of inputs and outputs. The control unit minimum number of required arc fault and thermal sensor inputs shall be as specified (see 6.2). The control unit minimum number of required trip signal outputs shall be as specified (see 6.2). The control unit minimum number of required alarm signal contacts shall be specified (see 6.2).

3.4.1.8.14 Accidental contact. The control unit shall provide protection to personnel against accidental contact with voltages in excess of 30 volts root mean square (RMS) or direct current (DC) with all panels in place and hinged doors closed. In addition, HMI access to program controls shall not expose the operator to voltages in excess of 30 volts RMS or DC.

3.4.1.8.15 Watch dog timer. Control units shall utilize a software watch dog timer circuit that shall automatically cycle power to the processor in the event of software ‘hanging’ or ‘freezing’.

3.4.1.9 Vibration. The control unit shall withstand the type I vibration requirements of MIL-STD-167-1 without loss of function, false alarms, uncommanded change in mode, loss of communications, other electrical malfunction or mechanical failure.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:29Z
Check the source to verify that this is the current version before use.

3.4.1.10 Shock. The control unit shall be in accordance with the grade A shock requirements of MIL-DTL-901.

3.4.1.11 Electromagnetic interference (EMI). The control unit shall be in accordance with the EMI limits and criteria of MIL-STD-461 for installation both below deck surface ships as well as installation aboard submarine.

3.4.1.12 DC magnetic field. The control unit shall withstand DC magnetic fields specified in DOD-STD-1399-70-1.

3.4.1.13 Insulation resistance. The control unit shall have an insulation resistance of not less than 10 megohms corrected to 25 °C when subjected to the insulation resistance methodology of MIL-STD-202-302 using the testing criteria specified in 4.4.3.

3.4.1.14 Dielectric strength. The control unit shall not show any detrimental effects when subjected to the dielectric withstanding voltage methodology of MIL-STD-202-302 using the testing criteria specified in 4.4.4.

3.4.1.15 Humidity. The control unit shall not show any detrimental effects or any accumulation of liquid water within the enclosure when subjected to the humidity methodology of MIL-STD-202-103 using the testing criteria specified in 4.4.14.

3.4.1.16 Inclined operation. The control unit shall not show any detrimental effects when subjected to the interface requirements and constraints of DOD-STD-1399-301 using the design limits for surface ships with sea state 8.

3.4.1.17 Sensor interface. The interface between the control unit and ship's cable to the network interface device (if used) or the sensors shall conform to the following:

a. Connector type. The connector used shall be a type MS3470W12-10SW, or equal.

b. Connector pinning. Connector pinning shall be as follows:

(1) Pin "A" shall be +12 VDC sensor power.

(2) Pin "B" shall be power ground.

(3) Pin "D" shall be RS485 negative.

(4) Pin "E" shall be RS485 positive.

(5) Pin "F" shall be chassis ground for network shield.

(6) Pin "G" shall be chassis ground for network shield.

(7) Pin "H" shall be chassis ground for network shield.

(8) Pin "J" shall be power ground.

(9) Pin "K" shall be +12 VDC sensor power.

c. Communication standard. Communication between the control unit and network interface equipment (if used) or the sensor shall be in accordance with TIA-485.

3.4.2 Photosensors. The photosensor shall be activated by visible and ultra-violet radiant energy from an electrical arc. Additional detailed photosensor requirements are called out on the applicable specification sheet.

3.4.2.1 Temperature ranges. The temperature ranges for the photosensor shall be as follows:

3.4.2.1.1 Operating temperature range. The photosensor shall operate in a temperature range of 0 to +75 °C.

3.4.2.1.2 Non-operating temperature range. The photosensor shall operate after storage in a temperature range of -40 to +75 °C.

3.4.2.2 Operational humidity range. The photosensor shall operate in humidity conditions of 5 to 95 percent relative humidity (RH) when subjected to the methodology of MIL-STD-202-103.

3.4.2.3 Maintainability. The photosensor shall not require servicing by the ship's crew other than for cleaning as needed. Cables and sensors are preferred to be independently repairable or replaceable.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:29Z
Check the source to verify that this is the current version before use.

3.4.2.4 Mounting. Provisions for mounting with screws shall be required. Alternative mounting may be used when specified (see 6.2).

3.4.2.5 Cable length. The photosensor shall have cable runs available of at least 30 meters (100 feet) to the control unit taking into account any intermediate devices.

3.4.2.6 Vibration. The photosensor shall withstand type I vibration requirements of MIL-STD-167-1 without mechanical damage, electrical damage, degradation of function, or activation of arc detection signal.

3.4.2.7 Shock. The photosensor shall be in accordance with the grade A shock requirements of MIL-DTL-901.

3.4.2.8 Photosensor function. The photosensor shall be in accordance with the applicable functional requirements of the applicable specification sheet.

3.4.3 CTMDs. Unless otherwise specified (see 6.2), the CTMD shall be in accordance with both the below requirements and the applicable functional requirements of the specification sheet.

3.4.3.1 Mounting. Provisions for mounting with screws or bolts shall be required. Alternative mounting may be used when specified (see 6.2).

3.4.3.2 Temperature ranges. The temperature ranges for the CTMD shall be as follows:

3.4.3.2.1 Operating temperature range. The CTMD shall operate in an ambient air temperature range of 0 to +75 °C.

3.4.3.2.2 Non-operating temperature range. The CTMD shall operate after storage in a temperature range of -40 to +75 °C.

3.4.3.3 Operational humidity range. The CTMD shall operate in humidity conditions of 5 to 95 percent RH when subjected to the methodology of MIL-STD-202-103.

3.4.3.4 Maintainability. The CTMD shall not require servicing by the ship’s crew other than for cleaning as needed. Cables and sensors are preferred to be independently repairable or replaceable.

3.4.3.5 Cable length. The CTMD shall have cable runs available of at least 30 meters (100 feet) to the control unit taking into account any intermediate devices.

3.4.3.6 Safety. If a CTMD contains a radioactive Americium-241 source in the ionization chamber, all federal, state, local, and United States Nuclear Regulatory Commission regulations shall be complied with applicable to the manufacture, importation, distribution, marking, and disposal of all items containing radioactive byproduct materials.

3.4.3.7 Markings. If a CTMD contains a radioactive Americium-241 source in the ionization chamber, the CTMD shall have United States Nuclear Regulatory Commission approval for distribution without restrictions and each unit shall contain approved markings identifying that the CTMD can be distributed without restrictions.

3.4.3.8 Vibration. The CTMD shall withstand the type I vibration requirements of MIL-STD-167-1 without mechanical damage, electrical damage, degradation of function, or activation of thermal detection signal.

3.4.3.9 Shock. The CTMD shall be in accordance with the grade A shock requirements of MIL-DTL-901.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First article inspection. First article inspection shall be performed on the AFPS when a first article sample is required (see 3.2). This inspection shall include the conformance inspection and the additional tests of 4.4.8 through 4.4.18.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:29Z
Check the source to verify that this is the current version before use.

4.2.1 First article system. The first article system shall be defined as the following: one control unit, sensors, associated cables, junction boxes, interfaces, and components as specified by the individual tests.

4.3 Conformance inspection. Conformance inspection shall include the tests listed in table I in the order shown.

TABLE I. Examinations and inspections.

Test requirement	Requirement	Verification	First article	Conformance
Materials of construction	3.3	4.4.1	X 1/	X 1/
Physical	3.4.1.2	4.4.1	X	X
Nameplates	3.4.1.4	4.4.1	X	X
HMI	3.4.1.8.2	4.4.1	X	X
Data log/archive file	3.4.1.8.4	4.4.1	X 1/	X 1/
Modes of operation	3.4.1.8.7	4.4.1	X	
Input power switch	3.4.1.8.9	4.4.1	X	X
Input power fuses	3.4.1.8.10	4.4.1	X	X
Restoring control unit and sensors	3.4.1.8.11	4.4.1	X 1/	X 1/
Baseline configurations	3.4.1.8.12	4.4.1	X 1/	X 1/
Minimum inputs and outputs	3.4.1.8.13	4.4.1	X	X
Watchdog timer	3.4.1.8.15	4.4.1	X 1/	X 1/
Sensor interface	3.4.1.17	4.4.1	X 1/	
Maintainability	3.4.2.3	4.4.1	X 1/	X 1/
Mounting	3.4.2.4	4.4.1	X	X
Cable length	3.4.2.5	4.4.1	X	X
CTMDs	3.4.3	4.4.1	X	X
Mounting	3.4.3.1	4.4.1	X	X
Maintainability	3.4.3.4	4.4.1	X	X
Cable length	3.4.3.5	4.4.1	X	X
Safety	3.4.3.6	4.4.1	X 1/	X 1/
Markings	3.4.3.7	4.4.1	X	X
Grounding	3.4.1.2.1	4.4.2	X	X
Insulation resistance	3.4.1.13	4.4.3	X	X
Dielectric strength	3.4.1.14	4.4.4	X	X
Control unit end-to-end test	3.4.1.5, 3.4.1.8.1, 3.4.1.8.3, 3.4.1.8.5, 3.4.1.8.7.1, and 3.4.1.8.8	4.4.5	X	X
Photosensor functional test	3.4.2 and 3.4.2.8	4.4.6	X	X
CTMD functional test	3.4.1.6	4.4.7	X	X
Inclined operation	3.4.1.16	4.4.9	X	

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:29Z
Check the source to verify that this is the current version before use.

TABLE I. Examinations and inspections - Continued.

Test requirement	Requirement	Verification	First article	Conformance
High temperature operation	3.4.1.3.1, 3.4.2.1.1, and 3.4.3.2.1	4.4.10	X	
Low temperature operation	3.4.1.3.1, 3.4.2.1.1, and 3.4.3.2.1	4.4.10.1	X	
Storage temperature	3.4.1.3.2, 3.4.2.1.2, and 3.4.3.2.2	4.4.10.2	X	
EMI	3.4.1.11	4.4.11	X	
DC magnetic field	3.4.1.12	4.4.12	X	
Shipboard power	3.4.1.7	4.4.13	X	
Humidity	3.4.1.15, 3.4.2.2, and 3.4.3.3	4.4.14	X	
Control unit enclosure	3.4.1.1	4.4.15	X	
Vibration	3.4.1.9, 3.4.2.6, and 3.4.3.8	4.4.16	X	
Shock	3.4.1.10, 3.4.2.7, and 3.4.3.9	4.4.17	X	
Software validation	3.4.1.8.6 and 3.4.1.8.8	4.4.18	X	

NOTE:

1/ Indicates a requirement that may be verified with appropriate documentary evidence (see 4.4.1).

4.4 Detailed inspections.

4.4.1 General examination. The completed AFPS unit shall be given a thorough examination to determine that it conforms to the applicable specifications and manufacturer's top level drawings with respect to safety, material, finish, workmanship, construction, assembly, electrical parameters and function of electrical components, dimensions, weight, and marking of identification and description plates. This examination shall be limited to those that can be performed without disassembling the unit in such a manner that its performance, durability, or appearance would be affected. Noncompliance with any general examination requirements or the presence of one or more defects shall constitute cause for rejection. Requirements verified by 4.4.1 in table I shall be visually examined. For requirements in table I that are software or documentary in nature, documentation shall be provided as specified (see 6.2).

4.4.2 Continuity test. A point-to-point verification continuity of all wired connections to applicable schematics and wiring drawings shall be performed. Noncompliance with any continuity test requirements shall constitute cause for rejection.

4.4.3 Insulation resistance test. The control unit and components shall be subjected to insulation resistance testing in accordance with MIL-STD-202-302. Test conditions shall be as follows:

a. Test potential: 500 volts ±10 percent for operating potentials of 450 VAC or less.

b. Points of measure: between each electrically isolated circuit and all other circuits connected together to ground (frame, chassis, or enclosure as applicable).

c. Application time of test voltage shall be sufficient to take resistance readings.

d. Temperature at time of test: temperature of parts to be tested shall be measured and recorded.

Insulation resistance shall be not less than 10 megohms corrected to 25 °C for any readings. Corrections shall be made on the basis of insulation resistance doubling for each 10 °C decrease in temperature.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:29Z
Check the source to verify that this is the current version before use.

4.4.4 Dielectric strength test. The AFPS control unit and components shall withstand a dielectric test in accordance with MIL-STD-202-301. Test conditions shall be as follows:

a. Magnitude of test voltage.

(1) For circuits rated 60 volts or less, the RMS test voltage shall be 900 volts.

(2) For circuits rated more than 60 volts, the RMS test voltage shall be twice-rated circuit voltage plus 1,000 volts.

(3) For circuits containing parts that are applied within their specified ratings and are in accordance with part specifications specified in 4.4.4.a(1) or (2), the dielectric test voltage for the circuit shall correspond to that specified for the approved part having the lowest specified dielectric test voltage, but in no case less than:

(a) 900 volts for circuits connected to the power supply terminals.

(b) 500 volts for circuits electrically isolated from the power supply terminals.

b. Nature of test voltage: the test voltage shall approximate a true sine wave of a frequency of 60 hertz.

c. Points of application: the test voltage shall be applied successively between each electrically isolated circuit and grounded with all other circuits grounded.

d. Duration of test: the duration of the test shall be not less than 60 seconds. The test voltage shall be removed by adjustment of its value to zero; not by sudden interruption.

Any evidence of arcing, flashover, odor, or punctured insulation shall be interpreted as a failure. Tripping of the test equipment shall also constitute failure.

4.4.5 Control unit end-to-end test. The control unit shall be subjected to an end-to-end test.

4.4.5.1 BIT. The control unit shall be subjected to a BIT test.

a. Frequency. The control unit shall be verified to be capable of automatically executing the BIT at a frequency not exceeding 24 hours. The BIT shall also be verified for capability of execution upon initiation by an operator at the HMI.

b. Components. The BIT shall be verified to test the control unit, the sensors, and the coil of the remote relay used to trip the circuit breaker. The BIT shall be verified to indicate a failure if the remote trip interface relay circuit is shorted or open. The BIT shall be verified to indicate a failure if any sensor circuit is open, shorted, or the sensor is otherwise nonfunctional.

c. Data log. Archive/log data of system configuration data, BIT operation, and BIT failure data shall be verified. All data shall be verified to be date- and time-stamped.

d. Additional acceptance criteria. Activation of the BIT shall be verified to not cause activation of remote trip interface relays or alarm outputs. Failure of a component during the BIT execution shall be verified to be identified at HMI; failure of a component during BIT execution shall be verified to not be excluded from the subsequent BIT.

4.4.5.2 Arc fault timing tests. Unless otherwise specified (see 6.2), it shall be demonstrated with actual or simulated arc flashes that the control unit shall ignore all arc flash signals of less than 45 milliseconds, which includes inputs from multiple point sensors. It shall be demonstrated that each sensor resets only after all sensors lose the light signal. Using an oscilloscope or equivalent for monitoring and using an actual or simulated arc flash of sufficient threshold, it shall be verified that the control unit sends a trip signal within a total time of that specified (see 6.2) from the start of the incoming arc flash signal.

4.4.5.3 Arc fault local/remote alarm test. The control unit shall be verified to sound an audible local alarm to indicate an arc fault when each photosensor is stimulated. Resistance shall be measured across each of the alarm pairs to verify 7,000 ohms ±5 percent in the non-alarm state and 0 ohms in the alarm state. Archive/log data, arc fault event data, and sensor(s) reporting data shall be verified. All data shall be verified to be date- and time-stamped. It shall be verified that alarms are capable of being reset at the local HMI.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:29Z
Check the source to verify that this is the current version before use.

4.4.5.4 Thermal local/remote alarm test. The control unit shall be verified to sound an audible local alarm to indicate a thermal event when each CTMD sensor is stimulated. Resistance shall be measured across each of the alarm pairs to verify 7,000 ohms ±5 percent in the non-alarm state and 0 ohms in the alarm state. Activation of the thermal monitoring alarm shall be verified to not cause activation of the breaker trip circuits. It shall be verified that alarms are capable of being reset at the local HMI. The method of testing shall be in accordance with MIL-DTL-32656/2.

4.4.5.5 Trip circuit. Each trip circuit shall demonstrate that it is capable of driving 113 milliamps at a sustained 115 VAC at a frequency of 60 hertz. When a control unit has multiple trip circuits serving a single monitoring point, all trip circuits shall demonstrate simultaneous activation.

4.4.5.6 Input power switch. The control unit power switch shall be demonstrated to operate a shut down of the control unit.

4.4.5.7 Communications. The control unit shall be demonstrated to communicate and interact with an external device when specified (see 6.2).

4.4.5.8 Disable compartments. When a control unit has the ability to monitor multiple compartments, the ability to shut down and reactivate all the sensors applicable to a specific compartment shall be demonstrated.

4.4.6 Photosensor functional test. Photosensors shall be subjected to functional testing as detailed in the applicable specification sheet.

4.4.7 Continuous thermal monitoring device functional test. The CTMD shall be subjected to functional testing to validate the accuracy of the alarm setting to detect overheated cable and bus connections in accordance with the applicable specification sheet.

4.4.8 Powered burn-in test. The first article control unit shall be subjected to a powered burn-in test. The 100-hour burn-in test shall be conducted by energizing 115 VAC voltage to the equipment for 100 hours at nominal voltage and frequency, ambient temperature, and with all inputs and outputs connected to effect maximum rated load during non-alarming conditions. The control unit shall be connected to the maximum number of sensors and sensor interrogation devices capable of being monitored by one control unit during this test or simulated loads of the devices. If a failure occurs during this test, the failure shall be corrected, and the test shall be repeated from the start. The control unit shall successfully pass the BIT after the test.

4.4.9 Inclined operation test. The control unit shall be tested for inclined operation to the levels specified in DOD-STD-1399-301 for the application specified (see 6.2). The AFPS shall be operational during the test and shall successfully complete a BIT before, during, and after the test.

4.4.10 High temperature operation test. The equipment under test shall be subjected to an operational temperature test in accordance with MIL-STD-810, method 501, procedure II, “Operation”. Temperature shall be held at the required high operation temperature for a period of no less than 2 hours after the temperature has stabilized, with the system operational. Any aberrant system behavior during the stabilization period, including but not limited to false alarms of either the CTMD or photosensors, communication errors, or a loss of system responsiveness shall be cause for failure. At the conclusion of the stabilization period, the system shall successfully complete a BIT. Photosensors shall be activated and the control unit shall send a trip signal and alarm while operating at high temperature. The CTMD shall be activated and the control unit shall alarm while operating at high temperature.

4.4.10.1 Low temperature operation test. The equipment under test shall be subjected to an operational temperature test in accordance with MIL-STD-810, method 502, procedure II, “Operation”. Temperature shall be held at the required low operation temperature for a period of no less than 2 hours after the temperature has stabilized, with the system operational. Any aberrant system behavior during the stabilization period, including but not limited to false alarms of either the CTMD or photosensors, communication errors, or a loss of system responsiveness shall be cause for failure. At the conclusion of the stabilization period, the system shall successfully complete a BIT. Photosensors shall be activated and the control unit shall send a trip signal and alarm while operating at low temperature. The CTMD shall be activated and the control unit shall alarm while operating at low temperature

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:29Z
Check the source to verify that this is the current version before use.

4.4.10.2 Temperature storage test. The control unit and components shall be subjected to a high temperature test in accordance with MIL-STD-810, method 501, procedure I, “Storage”. Exposure cycles shall be determined by the applicable requirements paragraph. Operational test post-exposure shall include successful boot-up, system BIT, and photosensor activation. Failure to boot, complete BIT, or successfully issue relay signal from photosensor activation shall be cause for test failure.

4.4.11 EMI test. The control unit shall be subjected to EMI tests in accordance with MIL-STD-461 and shall be conducted on electronic devices. Testing shall not cause false trip signals or failure of electronic devices. The control unit shall be operational during the tests and shall successfully complete a BIT before and after each test in the sequence. The control unit shall be connected to the maximum number of sensors and sensor interrogation devices (if used), capable of being monitored by one control unit during this test.

4.4.12 DC magnetic field test. The control unit shall be subjected to a DC magnetic field test in accordance with DOD-STD-1399-70-1. Testing shall not cause false trip signals or failure of electronic devices. The control unit shall be operational during the tests and shall successfully complete a BIT before and after each test. The control unit shall be connected to sensors during this test.

4.4.13 Shipboard power test. The control unit input power lines shall be subjected to the complete series of tests in accordance with MIL-STD-1399-300-1, type I power. Testing shall not cause false trip signals or failure of electronic devices. The control unit shall successfully pass a BIT test before and after each test. The control unit shall remain fully operational throughout the 70 milliseconds loss of power interruption test without loss of data or a requirement for a manual restart. The data log file shall be verified to be retained when tested to the 2-minute power interruption test.

4.4.14 Humidity test. The control unit shall be tested in accordance with MIL-STD-202-103 for a minimum of 96 hours to determine compliance with humidity requirements. The control unit shall be operational during the tests and shall successfully complete a BIT before and after each test. The control unit shall be connected to sensors and any additional components that are also included inside the test chamber during this test.

4.4.15 Degree of enclosure test. The control unit shall be tested in accordance with MIL-STD-108 to determine the effectiveness of the enclosure. The degree of enclosure shall be spraytight. The control unit shall be operational during the tests and shall successfully complete a BIT before and after each test. Additionally, the unit shall be tested for dielectric strength in accordance with 4.4.4 within 30 minutes after completion of the enclosure test.

4.4.16 Vibration test. The complete AFPS consisting of a control unit and components shall be subjected to the type I tests in accordance with MIL-STD-167-1. Unless otherwise specified (see 6.2), maximum frequency of vibration shall be in accordance with MIL-STD-167-1. The test setup shall be the same as specified in 4.4.17.5. The control unit shall successfully pass the BIT both prior to vibration testing and after completion of test. Failure to pass the post-vibration BIT shall constitute failure of the vibration test. Any mechanical or electrical damage, uncommanded change in operational state of control unit, or spurious activation of system shall constitute failure of vibration test.

4.4.17 High-impact shock test. The high-impact shock test shall be as specified in 4.4.17.1 to 4.4.17.4. Any additional shock ordering data shall be as specified (see 6.2).

4.4.17.1 High-impact shock (class I control unit). The control unit (principal unit) designed for hard mounting shall be subjected to grade A, class I, type A, high-impact shock testing in accordance with MIL-DTL-901. The control unit shall meet the shock test acceptance criteria of 4.4.17.6.

4.4.17.2 High-impact shock (class II control unit). The control unit (principal unit) designed for shock isolators shall be subjected to grade A, class II, type A, high-impact shock testing in accordance with MIL-DTL-901. The control unit shall meet the shock test acceptance criteria of 4.4.17.6.

4.4.17.3 High-impact shock (class I subassembly components). The components designed for hard-mounted enclosures (subassembly unit) shall be subjected to grade A, class I, type C, high-impact shock testing in accordance with MIL-DTL-901. All subassembly components installed shall meet the shock test acceptance criteria of 4.4.17.6.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:29Z
Check the source to verify that this is the current version before use.

4.4.17.4 High-impact shock (class II subassembly components). The components designed for shock-isolated enclosures (subassembly unit) shall be subjected to grade A, class II, type C, high-impact shock testing in accordance with MIL-DTL-901. All subassembly components installed shall meet the shock test acceptance criteria of 4.4.17.6.

4.4.17.5 Test setup. Test setup shall include the following:

a. All AFPS components consisting of a control unit and components shall be energized and operational during the test and shall successfully complete a BIT before the test. Only one example of each subassembly component needs to be tested.

b. Typical cable, cable terminations, and cable restraints required for shipboard operation shall be included in shock testing.

c. The AFPS shall be monitored for false trip signals sent to external multi-deck rotary relays using an oscilloscope or equivalent.

4.4.17.6 Test acceptance criteria. The following pass/fail criteria shall be used to determine AFPS component acceptance:

a. Hair line cracks are permitted in the control unit framework. Separation of parts shall constitute cause for test failure.

b. Control unit enclosure distortion, to an extent which might result in contact with any electrically conducting part within the enclosure, shall constitute cause for test failure.

c. False trip signals or failure of electronic devices shall constitute cause for test failure.

d. Insulation resistance testing (see 4.4.3) to verify insulation integrity shall be completed on all AFPS components after completion of all test sequences. Failure to maintain insulation integrity as specified in 4.4.3 shall constitute failure of shock test.

e. The BIT shall be executed manually after completion of each test event. BIT failure to start, failure to complete, or any component indicating failure during the BIT shall constitute failure of shock test.

4.4.18 Software validation. All user-accessible programmable commands controlled by software, such as time delay settings or retrieving data log information, shall be subjected to validation to show that they are fully functional over their range of operation and in conformance with the requirements of this specification. Data from previous tests may be submitted to show programmable commands are fully operational. The control unit shall be connected to the maximum number of sensors and sensor interrogation devices (if used), capable of being monitored by one control unit during this test and the full functionality of the software shall be demonstrated.

4.4.18.1 Cybersecurity validation. When remote communication is specified (see 6.2), all additional cybersecurity requirements of this specification shall be subject to validation to show that they are fully functional and in conformance with the requirements of this specification.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The AFPS is intended to process sensor data, determine if an arc fault or thermal event is occurring, send out signals to isolate power, and test the system components. Due to the unique operating environment of this item, a commercially-available product is not a viable alternative.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:29Z
Check the source to verify that this is the current version before use.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. When first article inspection is required (see 3.2).

c. Request for approval from the Technical Authority to use alternative materials (see 3.3).

d. Request for approval from the Technical Authority to use alternative creepage and clearance distances (see 3.3.2).

e. Request for approval from the Technical Authority to use alternative electrical connectors (see 3.3.4).

f. Requirement for CTMD protection (see 3.4, 3.4.c, and 3.4.3).

g. Requirement for CTMD type, if thermal ionization detector type is not desired (see 3.4.c).

h. Arc fault times (trip delay and total operating) (see 3.4.1.5 and 4.4.5.2).

i. Requirement for the control unit activating an audible local alarm that sounds when an over-temperature event is detected by the system (see 3.4.1.8.3.2).

j. Requirement for data storage (see 3.4.1.8.4).

k. Remote communications/communication protocol and functions when required (see 3.4.1.8.8, 4.4.5.7, and 4.4.18.1).

l. Requirement to provide procedures to restore the control unit and its associated sensors (see 3.4.1.8.11).

m. Minimum number of feature inputs and outputs (see 3.4.1.8.13).

n. Alternative mounting when not as specified (see 3.4.2.4 and 3.4.3.1).

o. Requirement to provide documentation of items that are software or documentary in nature (see 4.4.1).

p. Requirements specified for the control unit inclined operation test levels (see 4.4.9).

q. Maximum frequency of vibration if other than specified in MIL-STD-167-1 (see 4.4.16).

r. Additional shock ordering data (see 4.4.17).

s. Requirements for remote communications and additional cybersecurity requirements specified for secure operation (4.4.18.1).

t. Packaging requirements (see 5.1).

6.3 Definitions.

6.3.1 Cybersecurity. Cybersecurity is the prevention of damage to, protection of, and restoration of computers, electronic communication systems, electronic communication services, wire communication, and electronic communication, including information contained therein, to ensure its availability, integrity, authentication, confidentiality, and non-repudiation.

6.3.2 Technical authority. The Technical Authority has the authority, responsibility, and accountability to establish, monitor, and approve technical standards, tools, and processes in conformance to higher authority policy, requirements, architectures, and standards. NAVSEA 05S may be contacted to determine the appropriate Technical Authority for any given requirement related to equipment and systems of U.S. Navy ships and submarines.

6.4 Subject term (key word) listing.

Continuous thermal monitoring device

CTMD

Photosensor

Thermal ionization detector

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:29Z
Check the source to verify that this is the current version before use.

CONCLUDING MATERIAL

Custodians:
- Army – AV
- Navy – SH
- Air Force – 85
- DLA – GS

Review activities:
- Army – CR, MI
- Navy – AS, CG, YD
- Air Force – 03, 19, 84
- DLA – GS8

Civil agency:
- GSA – FAS

Preparing activity:
- Navy – SH

(Project 5975-2019-001)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:29Z
Check the source to verify that this is the current version before use.